

C615

# Physics: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A 1,200 kg car made by a company with plants on three continents has an engine force of 3,000 N forward and total friction and drag of 600 N backward. The driver has a spotless record. What is the car's acceleration?

- A.  $2.5 \text{ m/s}^2$
- B.  $2 \text{ m/s}^2$
- C.  $0.5 \text{ m/s}^2$
- D.  $3 \text{ m/s}^2$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

A 10,000 kg delivery truck owned by a regional grocery chain collides head-on with a 1,000 kg compact car. The truck was carrying frozen food. During the collision, how does the force on the car compare with the force on the truck?

- A. The truck exerts ten times more force.
- B. The car exerts more force because it is lighter.
- C. They cancel, so neither vehicle accelerates.
- D. They are equal in size and opposite in direction.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3. [TRANSFER — reversed stem / different variable]

A book rests on a table. What is the reaction force paired with Earth's gravitational pull on the book?

- A. The book's downward push on the table
- B. The table's upward push on the book
- C. The book's gravitational pull upward on Earth
- D. None; gravity has no reaction force

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A 500 kg roller coaster car at a park that opened in 1985 starts from rest and drops 20 m down a frictionless track. The ride has a 4-star rating. Using  $g = 9.8 \text{ m/s}^2$ , what is its speed at the bottom?

- A. About 19.8 m/s
- B. 14 m/s
- C. About 0.89 m/s
- D. 392 m/s

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ____   |                               |

5. [TRANSFER — reversed stem / different variable]

A ball is thrown straight up at 10 m/s. Using  $g = 10 \text{ m/s}^2$  and no air resistance, how high does it rise?

- A. 10 m
- B. 1 m
- C. 50 m
- D. 5 m

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ____   |                               |

6.

On a lab track bought with a district grant, a 2 kg cart moving at 3 m/s strikes a 1 kg cart at rest, and they stick together. The lab has a new air-conditioning unit. What is their speed after the collision?

- A. 3 m/s
- B. 2 m/s
- C. 1.5 m/s
- D. 6 m/s

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ____   |                               |

7.

A science teacher who builds kits for a summer camp connects two  $6\ \Omega$  resistors in parallel to a 12 V battery. The kits come in reusable plastic bins. What current flows from the battery?

- A. 4 A
- B. 1 A
- C. 2 A
- D. 36 A

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8. [TRANSFER — reversed stem / different variable]

A third resistor is added in parallel to a circuit powered by a fixed-voltage battery. What happens to the total current from the battery?

- A. It increases.
- B. It stays the same because voltage is fixed.
- C. It always exactly doubles.
- D. It decreases.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

A band teacher at a school with a 400-seat auditorium plays a 170 Hz note. Sound travels at 340 m/s in the room, which was just re-carpeted. What is the wavelength?

- A. 340 m
- B. 2 m
- C. 0.5 m
- D. 57,800 m

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

At the same instant, a 0.2 kg ball is dropped from a 5 m ledge while a 0.5 kg ball is launched horizontally at 8 m/s from the same height. The ledge is part of a physics building with a green roof. Ignoring air resistance, which lands first?

- A. The launched ball lands later because it travels farther.
- B. The launched ball lands first because it is moving faster.
- C. They land at the same time.
- D. The heavier ball lands first.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.** [TRANSFER — reversed stem / different variable]

A ball rolls off a 20 m cliff at 5 m/s horizontally. Using  $g = 10 \text{ m/s}^2$ , how far from the base does it land?

- A. 20 m
- B. 100 m
- C. 10 m
- D. 5 m

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.**

Two small charged spheres in a lab funded by a national agency are moved from 1 cm apart to 3 cm apart. The lab keeps humidity at 40%. How does the electric force between them change?

- A. It stays the same because the charges did not change.
- B. It becomes 1/3 as large.
- C. It becomes 9 times as large.
- D. It becomes 1/9 as large.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer B Concept 1

A 1,200 kg car ~~made by a company with plants on three continents~~ has an engine force of 3,000 N forward and total friction and drag of 600 N backward. ~~The driver has a spotless record.~~ What is the car's acceleration?

**Target word:** acceleration from net force **Decides:** Net force = 3,000 – 600 = 2,400 N.

|          |                    |                                        |
|----------|--------------------|----------------------------------------|
| A        | Dropped element    | Ignores friction: $3,000 \div 1,200$ . |
| <b>B</b> | <b>Correct</b>     | $2,400 \div 1,200 = 2$ .               |
| C        | Confusion set      | Divides mass by force.                 |
| D        | Inverted operation | Adds friction to the engine force.     |

## 2. Answer D Concept 2

A 10,000 kg delivery truck ~~owned by a regional grocery chain~~ collides head-on with a 1,000 kg compact car. ~~The truck was carrying frozen food.~~ During the collision, how does the force on the car compare with the force on the truck?

**Target word:** force comparison **Decides:** Third-law pair.

|          |                    |                                                             |
|----------|--------------------|-------------------------------------------------------------|
| A        | Confusion set      | Mass affects acceleration, not the size of the pair forces. |
| B        | Inverted operation | Reverses the mass misconception; forces are still equal.    |
| C        | Adjacent category  | The forces act on different objects and cannot cancel.      |
| <b>D</b> | <b>Correct</b>     | Action-reaction forces are always equal.                    |

## 3. Answer C Concept 2 · transfer

A book rests on a table. What is the reaction force paired with Earth's gravitational pull on the book?

**Target word:** identify the third-law partner **Decides:** Swap the objects: book on Earth.

|          |                   |                                                              |
|----------|-------------------|--------------------------------------------------------------|
| A        | Adjacent category | Pairs with the table's push, not with gravity.               |
| B        | Confusion set     | Balances gravity on the book but is a different interaction. |
| <b>C</b> | <b>Correct</b>    | Same interaction, objects reversed.                          |
| D        | Absolute language | Every force has a third-law partner.                         |

## 4. Answer A Concept 3

A ~~500 kg~~ roller coaster car ~~at a park that opened in 1985~~ starts from rest and drops 20 m down a frictionless track. ~~The ride has a 4-star rating.~~ Using  $g = 9.8 \text{ m/s}^2$ , what is its speed at the bottom?

**Target word:** speed after the drop **Decides:**  $v = \sqrt{2gh}$ ; mass cancels.

|          |                    |                                                             |
|----------|--------------------|-------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | $\sqrt{2 \times 9.8 \times 20} = \sqrt{392} \approx 19.8$ . |
| B        | Dropped element    | Drops the factor 2: $\sqrt{9.8 \times 20}$ .                |
| C        | Camouflage         | Divides by the 500 kg mass, which cancels out.              |
| D        | Inverted operation | Forgets to take the square root.                            |



### 5. Answer D Concept 3 • transfer

A ball is thrown straight up at 10 m/s. Using  $g = 10 \text{ m/s}^2$  and no air resistance, how high does it rise?

**Target word:** solve for height **Decides:**  $h = v^2/(2g)$ .

|   |                   |                                               |
|---|-------------------|-----------------------------------------------|
| A | Dropped element   | Drops the factor 2.                           |
| B | Confusion set     | $v/g = 1$ is the time to the top, in seconds. |
| C | Wrong stem number | $v^2/2$ without dividing by $g$ .             |
| D | <b>Correct</b>    | $100 \div 20 = 5$ .                           |

### 6. Answer B Concept 4

On a lab track ~~bought with a district grant~~, a 2 kg cart moving at 3 m/s strikes a 1 kg cart at rest, and they stick together. ~~The lab has a new air conditioning unit.~~ What is their speed after the collision?

**Target word:** final speed, inelastic **Decides:** 6 kg·m/s shared by 3 kg.

|   |                   |                                      |
|---|-------------------|--------------------------------------|
| A | Dropped element   | Ignores the added mass.              |
| B | <b>Correct</b>    | $(2 \times 3) \div (2 + 1) = 2$ .    |
| C | Confusion set     | Averages the speeds 3 and 0.         |
| D | Wrong stem number | That is the momentum, not the speed. |

### 7. Answer A Concept 5

A science teacher ~~who builds kits for a summer camp~~ connects two 6  $\Omega$  resistors in parallel to a 12 V battery. ~~The kits come in reusable plastic bins.~~ What current flows from the battery?

**Target word:** total current **Decides:** Parallel 6  $\Omega$  and 6  $\Omega = 3 \Omega$ .

|   |                    |                                              |
|---|--------------------|----------------------------------------------|
| A | <b>Correct</b>     | $12 \text{ V} \div 3 \Omega = 4 \text{ A}$ . |
| B | Confusion set      | Adds them as series: $12 \div 12$ .          |
| C | Dropped element    | Current through one branch only.             |
| D | Inverted operation | Multiplies $V \times R$ instead of dividing. |

### 8. Answer A Concept 5 • transfer

A third resistor is added in parallel to a circuit powered by a fixed-voltage battery. What happens to the total current from the battery?

**Target word:** effect of adding a parallel branch **Decides:** Total resistance drops.

|   |                    |                                                           |
|---|--------------------|-----------------------------------------------------------|
| A | <b>Correct</b>     | Another path lowers total resistance, so $I = V/R$ rises. |
| B | Dropped element    | Drops the drop in total resistance.                       |
| C | Absolute language  | The increase depends on the resistor values.              |
| D | Inverted operation | Treats it like adding a series resistor.                  |

## 9. Answer B Concept 6

A band teacher at a school with a 400-seat auditorium plays a 170 Hz note. Sound travels at 340 m/s in the room, which was just re-carpeted. What is the wavelength?

**Target word:** wavelength **Decides:**  $\lambda = v / f$ .

|          |                    |                                      |
|----------|--------------------|--------------------------------------|
| A        | Wrong stem number  | Reports the speed as the wavelength. |
| <b>B</b> | <b>Correct</b>     | $340 \div 170 = 2$ .                 |
| C        | Confusion set      | Divides $f$ by $v$ .                 |
| D        | Inverted operation | Multiplies $v \times f$ .            |

## 10. Answer C Concept 7

At the same instant, a 0.2 kg ball is dropped from a 5 m ledge while a 0.5 kg ball is launched horizontally at 8 m/s from the same height. The ledge is part of a physics building with a green roof. Ignoring air resistance, which lands first?

**Target word:** which lands first **Decides:** Fall time depends only on height.

|          |                    |                                                              |
|----------|--------------------|--------------------------------------------------------------|
| A        | Confusion set      | Horizontal distance does not affect fall time.               |
| B        | Inverted operation | Horizontal speed has no effect on vertical fall.             |
| <b>C</b> | <b>Correct</b>     | Both start with zero vertical velocity from the same height. |
| D        | Camouflage         | Mass does not change fall time without air resistance.       |

## 11. Answer C Concept 7 • transfer

A ball rolls off a 20 m cliff at 5 m/s horizontally. Using  $g = 10 \text{ m/s}^2$ , how far from the base does it land?

**Target word:** horizontal distance **Decides:**  $t = \sqrt{(2h/g)} = 2 \text{ s}$ .

|          |                    |                                                     |
|----------|--------------------|-----------------------------------------------------|
| A        | Wrong stem number  | Reports the cliff height.                           |
| B        | Inverted operation | Multiplies speed by height.                         |
| <b>C</b> | <b>Correct</b>     | $5 \text{ m/s} \times 2 \text{ s} = 10 \text{ m}$ . |
| D        | Dropped element    | Uses 1 s of flight.                                 |

## 12. Answer D Concept 8

Two small charged spheres in a lab funded by a national agency are moved from 1 cm apart to 3 cm apart. The lab keeps humidity at 40%. How does the electric force between them change?

**Target word:** force change with distance **Decides:** Inverse square:  $1/3^2 = 1/9$ .

|          |                    |                                        |
|----------|--------------------|----------------------------------------|
| A        | Dropped element    | Drops the distance dependence.         |
| B        | Confusion set      | Uses $1/r$ instead of $1/r^2$ .        |
| C        | Inverted operation | Force falls, not rises, with distance. |
| <b>D</b> | <b>Correct</b>     | $F \propto 1/r^2$ .                    |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 8             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 4             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 2             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 9             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 2             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 2             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                     | Rule                                                                                                                                         | Trap                                                                                       | Tell                                                                 |
|---|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1 | <b>Newton's second law uses net force</b>   | $a = F_{\text{net}} / m$ : subtract opposing forces before dividing by mass.                                                                 | Dividing one applied force by mass while ignoring friction or drag.                        | Several forces acting along the same line.                           |
| 2 | <b>Newton's third law pairs</b>             | Action-reaction forces are equal in size, opposite in direction, and act on different objects, so they never cancel each other.              | Believing the larger or faster object exerts the larger force.                             | A collision between unequal objects or "what is the reaction force?" |
| 3 | <b>Conservation of mechanical energy</b>    | Without friction, $mgh = \frac{1}{2}mv^2$ , so speed after a drop is $\sqrt{2gh}$ and does not depend on mass.                               | Thinking heavier objects reach higher speeds, or dropping the square root or the factor 2. | A height and "how fast at the bottom?"                               |
| 4 | <b>Conservation of momentum</b>             | Total momentum before a collision equals total momentum after; in a perfectly inelastic collision, $v = (m_1v_1 + m_2v_2)/(m_1 + m_2)$ .     | Averaging speeds or ignoring the added mass.                                               | Objects that stick together.                                         |
| 5 | <b>Circuits: Ohm's law and combinations</b> | $V = IR$ ; series resistances add, while parallel branches combine as $1/R = 1/R_1 + 1/R_2$ , lowering total resistance.                     | Adding parallel resistors like series ones.                                                | A battery with resistors "in parallel".                              |
| 6 | <b>Wave speed relation</b>                  | $v = f\lambda$ ; the medium sets speed, so $\lambda = v/f$ .                                                                                 | Multiplying when solving for wavelength, or inverting the ratio.                           | A wave speed and a frequency.                                        |
| 7 | <b>Projectile independence</b>              | Horizontal and vertical motions are independent; fall time depends only on height, and horizontal distance = horizontal speed $\times$ time. | Believing a horizontally launched object stays up longer.                                  | One object dropped and another launched horizontally.                |
| 8 | <b>Inverse-square force laws</b>            | Coulomb and gravitational forces vary as $1/r^2$ ; tripling distance cuts force to $1/9$ .                                                   | Scaling force by $1/r$ instead of $1/r^2$ .                                                | "The distance is doubled/tripled."                                   |

**Course scope (public description):** Physics: Content Knowledge covers the advanced content knowledge that a secondary physics teacher is expected to know and understand. Topics include nature and impact of science and engineering, principle and models of matter and energy, mechanics, electricity and magnetism, waves, and science teaching and pedagogy.